(1)$\alpha+\beta=2,\ \alpha\beta=3,$　(商)$x+1,$ (余り)$x+2$　　(2)$\alpha^2+\beta^2=-2,$　$\{f(\alpha)\}^2+\{f(\beta)\}^2=14$

$(1) A(1,2), \quad l_1 : x + 2y = 5 \qquad (2) l_2 : 11x + 2y = 25, \quad C\left(\frac{11}{5}, \frac{2}{5}\right) \qquad (3) r = \frac{5}{6}, \quad \left(\frac{3}{2}, \frac{4}{3}\right)$

(1)$\overrightarrow{OD} = \frac{1}{4}\overrightarrow{a} + \frac{3}{4}\overrightarrow{b}$, $\overrightarrow{OM} = \frac{1}{8}\overrightarrow{a} + \frac{3}{8}\overrightarrow{b} + \frac{1}{8}\overrightarrow{c}$ (2)$\overrightarrow{OP} = \left(1 - \frac{7}{8}k\right)\overrightarrow{a} + \frac{3}{8}k\overrightarrow{b} + \frac{1}{8}k\overrightarrow{c}$, $k = \frac{8}{7}$ (3)$\overrightarrow{OH} = \frac{2}{7}\overrightarrow{a}$

(1)$b=12a$　　(2)(極大値)$12a-8$,　(極小値)$-a^3+6a^2$　　(3)$M=\begin{cases}32 & \left(2<a\leq\frac{10}{3}\right)\\ 12a-8 & \left(a\geq\frac{10}{3}\right)\end{cases}$

$(1) y=8$ $(2) x<1, \quad y=2-2x$ $(3) -\frac{9}{8}<k<0$